CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 12: **Magnetic Effects of Electric Current** | Quick revision, one page

Magnetic field and field lines

- Field lines: closed curves, from N to S outside the magnet and S to N inside; closer lines mean a stronger field; they never intersect; the tangent gives the field direction.
- Oersted: current in a wire deflects a compass needle (current produces a magnetic field). Field around a straight wire: concentric circles; strength increases with current and decreases with distance.
- Right-hand thumb rule: thumb along the current, curled fingers give the field direction. A circular loop: field at the centre is strongest, increases with the number of turns.

Solenoid and force

- Solenoid: a coil of many circular turns; field lines like those of a bar magnet; field inside is uniform; an iron core makes an electromagnet.
- Force on a conductor: maximum when the current is perpendicular to the field. Fleming's left-hand rule: forefinger = field, middle finger = current, thumb = force (mutually perpendicular).
- Reversing the current or the field reverses the force. Unit of field strength: tesla (T).

Electromagnetic induction

- A changing magnetic field through a coil (or a conductor moving in a field) induces an emf and a current.
- Fleming's right-hand rule: thumb = motion, forefinger = field, middle finger = induced current. Faster change or more turns gives greater induced current.
- Generator: converts mechanical energy into electrical energy (rotating coil in a magnetic field). AC reverses direction periodically (India: 50 Hz, i.e. reversal every $\frac{1}{100}$ s); DC flows in one direction.

Domestic electric circuits

- Supply 220 V in India: live wire (red), neutral (black), earth (green).
- Fuse: low melting point wire in the live wire; melts on overloading or short circuit. Earthing prevents shocks from the metal body. Appliances in parallel.
- Overloading: too many appliances on one circuit. Short circuit: live and neutral wires touch, resistance falls and current rises. $I = \frac{P}{V}$.